Dr. AMBEDKAR INSTITUTE OF TECHNOLOGY

Near Jnana Bharathi Campus, Bengaluru-560 056.

(An Autonomous Institution, Aided by Government of Karnataka)

Aided By Govt. of Karnataka

Project Report

on

"Income Tax Management system"

Submitted By

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Under the Guidance

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2024-2025

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CERTIFICATE

This is to certify that the project entitled "**Income Tax Management** " submitted in the partial fulfillment of the requirement of the 4th semester DBA laboratory curriculum during the year 2024-25 is a result of bonafide work carried out by

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USN:1DA23CS203

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Signature of the guides:

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Dr. AIT

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Assistant Professor, Dept of CSE
Dr. AIT

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H.O.D
Department of CSE, Dr. AIT

ACKNOWLEDGEMENT

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Finally, yet importantly, we would like to express our heartfelt thanks to our beloved **Parents** for their blessings and our **Friends** for their help and wishes for the successful completion of this project report.

Name1:Kavya M Moodi

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ABSTRACT

The Income Tax Management System (ITMS) is a web-based platform designed to simplify and streamline the tax filing process for individuals and organizations. By integrating taxpayer data management, professional collaboration, and automated notifications into a user-friendly interface, ITMS reduces errors, enhances transparency, and improves user experience. The platform features a personalized user dashboard for document uploads and tax status tracking, seamless integration with certified tax professionals, real-time notifications, and secure document management.

Built with a robust three-tier architecture comprising a responsive frontend (HTML, CSS), a PHP-driven backend for business logic, and a MySQL database for data storage, ITMS ensures scalability and maintainability. The system adheres to strict security practices to safeguard sensitive user information while enabling administrators to manage accounts and monitor system activity. This project embodies systematic software engineering principles, focusing on user-centric design and technological efficiency.

Table of Contents

Sl. No.	Topic	Page No
1.	Introduction	6
2.	Requirements Specification 2.1 Purpose of the Project 2.2 Scope of the Project 2.3 Hardware Requirements 2.4 Software Requirements	7
3.	Description 3.1 Explanation 3.2 Triggers and Procedures 3.3 Stored Procedure: AddTaxPr 3.4 ER Diagram 3.5 Relational Schema	9
4.	Code	14
5.	Snapshot	16
6.	Conclusion	27
7.	Bibliography	28

Chapter 1 Introduction

The growing complexity and scale of income tax administration in modern economies demand efficient, secure, and automated systems. Manual processes in tax filing, processing, and refund issuance not only lead to delays but are also prone to errors, fraud, and lack of transparency. In this context, the Income Tax Management System (ITMS) has been developed as a comprehensive solution to digitize and streamline the complete lifecycle of income tax operations.

The ITMS is designed to cater to three main user groups: Taxpayers, Tax Professionals, and Tax Authorities. It provides each stakeholder with a secure and role-based access platform to perform actions such as filing tax returns, uploading documents, calculating tax liabilities, processing payments and refunds, and generating reports. The system ensures data integrity and traceability through features like stored procedures, triggers, audit logs, and notification alerts.

By integrating a centralized database, rule-based automation, and user-friendly interfaces, the ITMS aims to reduce the administrative burden, enhance compliance, and provide real-time monitoring capabilities. The use of complex SQL queries enables advanced analytics, helping authorities make data-driven decisions and track taxpayer behavior.

Chapter 2 Requirement specification

2.1.Purpose of the Project

The Tax Management System (TMS) is designed to streamline tax-related operations for taxpayers, tax professionals, and tax authorities. This system aims to digitize tax processes, enabling secure and efficient management of revenue declarations, document uploads, and notifications. By integrating functionalities for all roles involved, TMS reduces manual paperwork and ensures seamless collaboration among stakeholders, ultimately enhancing transparency and accuracy in tax administration.

2.2. Scope of the Project

The scope of this project encompasses the following areas:

- **User Roles and Accessibility**: Three distinct roles—Taxpayer, TaxProfessional, and TaxAuthority—each with customized functionalities, ensuring role-based access control and data security.
- **Centralized Data Management**: Unified storage and retrieval of user information, tax documents, and notifications in a single database system.
- **Automated Operations**: Trigger-based automatic updates to maintain data integrity, such as the deletion of user records upon the removal of tax professionals.
- **Scalability**: The system is scalable to accommodate a growing number of users and extended functionalities, such as adding new modules for advanced tax analytics or compliance tracking.

2.3. Hardware Requirement

Component Minimum Requirement

Processor: IntelCore i5 or AMD Ryzen 5 (or higher)

RAM: 8 GB (16 GB recommended for smoother multitasking)

Hard Disk :500 GB HDD or 256 GB SSD

Monitor: 15.6-inch display with 1366x768 resolution (or higher)

Input Devices: Standard Keyboard and Mouse

Network Ethernet or Wi-Fi Adapter (for internet connectivity)

UPS Optional (for power backup during development/testing)

2.4. Software Requirements

Category	Name	Purpose
Programming Languages	Python	Backend development, automation, and scripting.
	PHP	Server-side scripting and database interaction.
	SQL	Database management and querying.
Tools	Selenium	Automated testing of web applications.
	phpMyAdmin	Database administration and management.
	MySQL Workbench	Designing and managing the database schema.
Software	Apache Server	Hosting the web application.
	XAMPP	Local server environment for development.
	Visual Studio Code	Code editing and debugging.

Chapter 3 Description

The **Tax Management System** is a web-based application that integrates the following features for its three user roles:

3.1.Explanation

1. **Common Features for All Users**:
   - A unified **login and signup page** for secure user authentication.
   - Centralized access to the portal based on role.
2. **Taxpayer Functionality**:
   - **Revenue Declaration**: Allows taxpayers to input their revenue details for tax calculations.
   - **Document Upload**: Taxpayers can upload relevant documents required for tax filing or audits.
   - **Notification Management**: View notifications from their assigned tax professionals for updates, reminders, or clarifications.
3. **TaxProfessional Functionality**:
   - **Taxpayer Management**: View a list of all assigned taxpayers.
   - **Detailed Taxpayer Information**: Access individual taxpayers' revenue data and uploaded documents, including recent and historical files.
   - **Notification System**: Send personalized notifications to taxpayers for communication.
4. **TaxAuthority Functionality**:
   - **Database Management**: View a joined table showing which taxpayers are assigned to which tax professionals.
   - **Add/Delete Tax Professionals**: Manage tax professionals in the system, including creating or removing their records.
5. **Technical Features**:
   - **Triggers and Procedures**:
     - Trigger for cascading user deletion when a tax professional is removed.

        - Procedure for efficiently adding tax professionals by automating entries in both USERand TaxProfessional tables.
    - **Role-Based Data Handling**: Different permissions and views based on roles to ensure data security and relevance.
6. **ER Diagram and Data Flow**:
    - The database is structured with normalized tables linking taxpayers, tax professionals, and tax authorities via foreign key relationships.
    - Documents and revenue data are linked to taxpayers, while tax professionals are linked to their respective users and certifications.

By integrating these functionalities, the Tax Management System addresses the critical needs of tax administration and simplifies workflows for all stakeholders.

3.2. Triggers and procedures:

Trigger: delete_user_on_taxprofessional

1. **Purpose**: Automatically deletes the associated user from the USER and Taxprofessionaltable when a TaxProfessionalrecord is deleted.
2. **Trigger Type**: AFTER DELETEtrigger, which executes after a TaxProfessionalrecord is removed.
3. **Functionality**:
    - Captures the user_idof the deleted TaxProfessionalusing the OLD keyword.
    - Deletes the corresponding record from the USERtable based on the captured user_id.
4. **Advantage**: Ensures data integrity by preventing orphaned user records when a tax professional is removed.

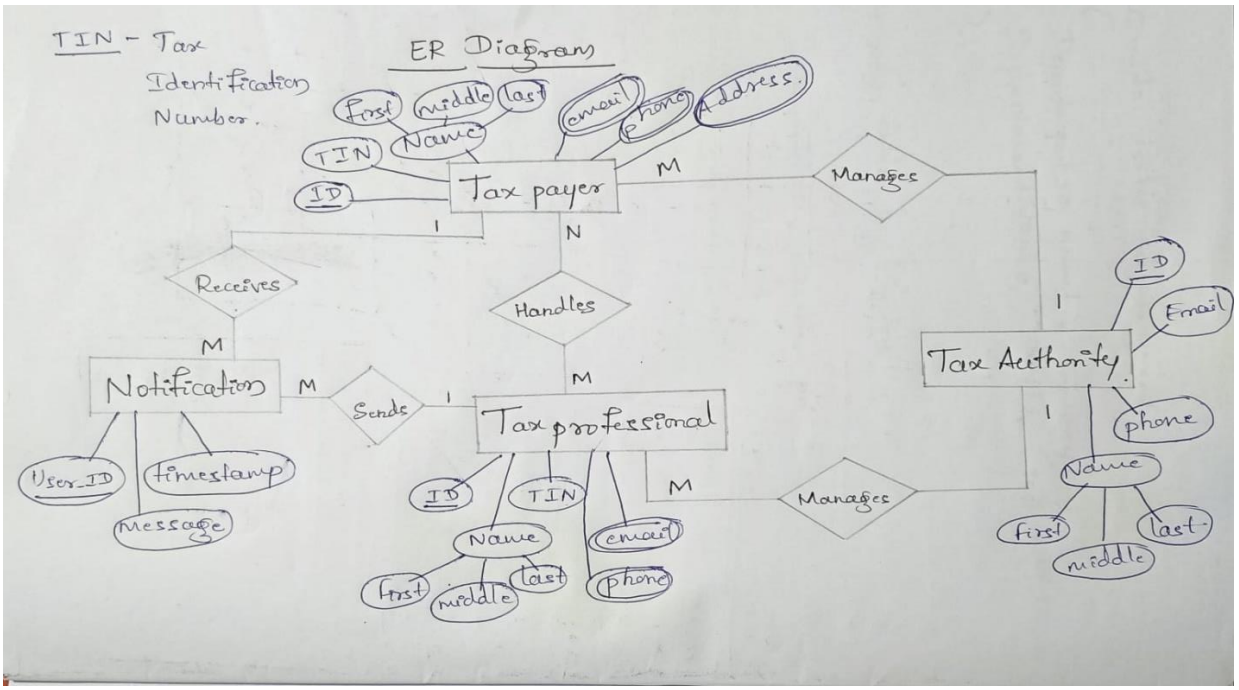
3.3.Stored Procedure: AddTaxProfessional

1. **Purpose**: Simplifies the process of adding a new tax professional by handling insertions into both the USERand TaxProfessionaltables.
2. **Parameters**:
    - Accepts details such as the name, email, phone number, TIN, password, and certification_idof the new tax professional.
3. **Functionality**:
    - Inserts the professional's details into the USERtable with the role set to 'TaxProfessional'.
    - Retrieves the user_idof the newly inserted user using LAST_INSERT_ID().
    - Inserts additional professional-specific details into the TaxProfessional table, including their TIN, certification, and registration date.
4. **Advantage**: Automates and streamlines the addition of tax professionals, reducing manual effort and potential errors.

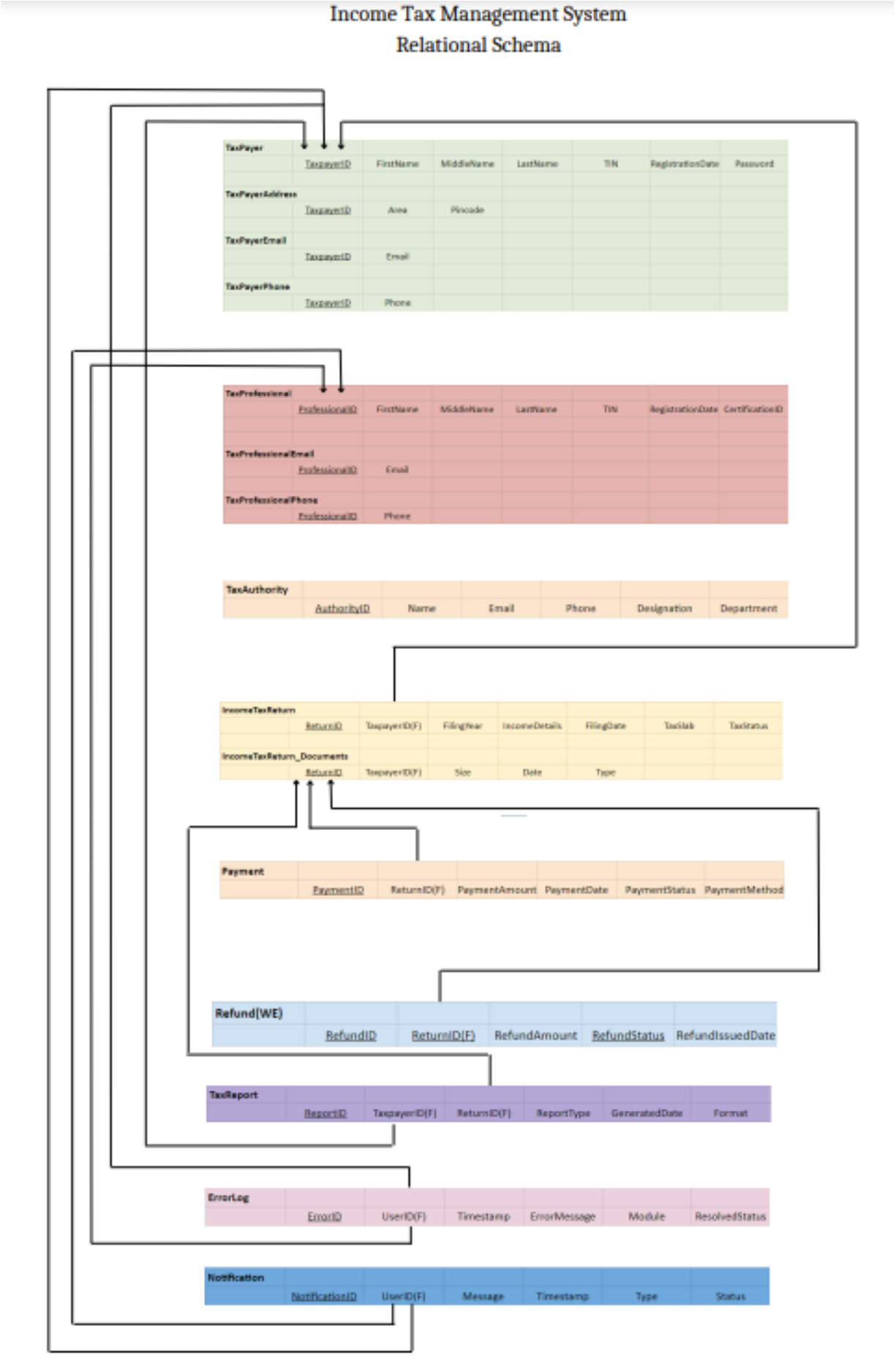
Join:

Join used in Taxauthority Dashboard where we joined two tables to get the combined table which shows the Taxprofessionals with their associated Taxpayers.

3.4. ER DIAGRAM



3.5. SCHEMA


Income Tax Management System
Relational Schema
TaxPayer
TaxpayerID
FirstName
MiddleName
LastName
TIN
RegistrationDate
Password
TaxPayerAddress
TaxpayerID
Area
Pincode
TaxPayerEmail
TaxpayerID
Email
TaxPayerPhone
TaxpayerID
Phone
TaxProfessional
ProfessionalID
FirstName
MiddleName
LastName
TIN
RegistrationDate
CertificationID
TaxProfessionalEmail
ProfessionalID
Email
TaxProfessionalPhone
ProfessionalID
Phone
TaxAuthority
AuthorityID
Name
Email
Phone
Designation
Department
IncomeTaxReturn
ReturnID
TaxpayerID(F)
FilingYear
IncomeDetails
FilingDate
TaxSlab
TaxStatus
IncomeTaxReturn_Documents
ReturnID
TaxpayerID(F)
Size
Date
Type
Payment
PaymentID
ReturnID(F)
PaymentAmount
PaymentDate
PaymentStatus
PaymentMethod
Refund(WE)
RefundID
ReturnID(F)
RefundAmount
RefundStatus
RefundIssuedDate
TaxReport
ReportID
TaxpayerID(F)
ReturnID(F)
ReportType
GeneratedDate
Format
ErrorLog
ErrorID
UserID(F)
Timestamp
ErrorMessage
Module
ResolvedStatus
Notification
NotificationID
UserID(F)
Message
Timestamp
Type
Status

Chapter 4 Coding

```
CREATE DATABASE IF NOT EXISTS itms;

USE itms;

CREATE TABLE User (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    password VARCHAR(255) NOT NULL,
    role ENUM('taxpayer', 'taxprofessional', 'taxauthority') NOT NULL
);


CREATE TABLE Taxpayer (
    TaxpayerID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100),
    TIN VARCHAR(15) UNIQUE,
    Address TEXT,
    Email VARCHAR(100) UNIQUE,
    Phone VARCHAR(15),
    RegistrationDate DATE,
    Password VARCHAR(255)
);

CREATE TABLE TaxProfessional (
    ProfessionalID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100),
    TIN VARCHAR(15),
    Certification_ID VARCHAR(20),
    Email VARCHAR(100) UNIQUE,
    Phone VARCHAR(15),
    RegistrationDate DATE
    Password VARCHAR(255)
);
```

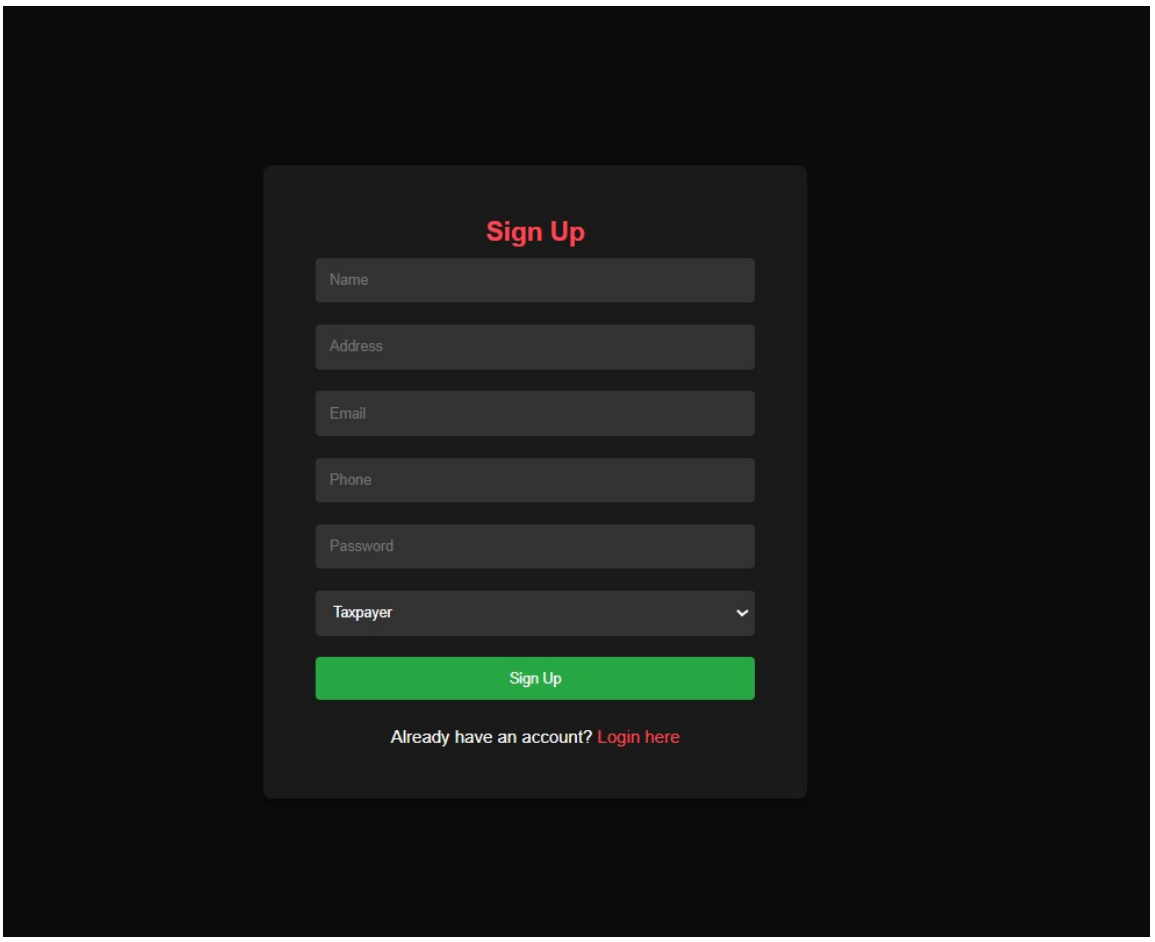
```sql
CREATE TABLE TaxAuthority (
    AuthorityID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100),
    Email VARCHAR(100) UNIQUE,
    Phone VARCHAR(15),
    Designation VARCHAR(50),
    Department VARCHAR(100)
);


CREATE TABLE Notification (
    NotificationID INT PRIMARY KEY AUTO_INCREMENT,
    UserID INT,
    Message TEXT,
    Timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    Type ENUM('Info', 'Warning', 'Alert'),
    Status ENUM('Unread', 'Read'),
    FOREIGN KEY (UserID) REFERENCES Taxpayer(TaxpayerID)
);
```

```php
<?php
include 'db.php';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $client_id = $_POST['client_id'];
    $revenue = $_POST['revenue'];

    // Define tax slabs
    $tax = 0;
    if ($revenue <= 250000) {
        $tax = 0; // No tax for income <= 2,50,000
    } elseif ($revenue <= 500000) {
        $tax = ($revenue - 250000) * 0.05;
    } elseif ($revenue <= 1000000) {
        $tax = 12500 + ($revenue - 500000) * 0.2;
    } else {
        $tax = 112500 + ($revenue - 1000000) * 0.3;
    }

    // Redirect back to the dashboard with the calculated tax
    header("Location: taxprofessional_dashboard.php?calculated_tax=" . urlencode($tax) . "&client_id=" . urlencode($client_id));
    exit();
}
?>
<?php if ($late_filings->num_rows > 0): ?>
    <div class="notification-box">
        <h3>⚠️ Late Filing Alerts</h3>
        <ul>
            <?php while ($row = $late_filings->fetch_assoc(): ?>
                <li>
                    Filing Year: <?php echo htmlspecialchars($row['FilingYear']); ?> –
                    Filed on: <?php echo htmlspecialchars($row['FilingDate']); ?>
                </li>
            <?php endwhile; ?>
        </ul>
    </div>
<?php endif; ?>
```

Chapter 5 Snapshots

Create:

(Signup)

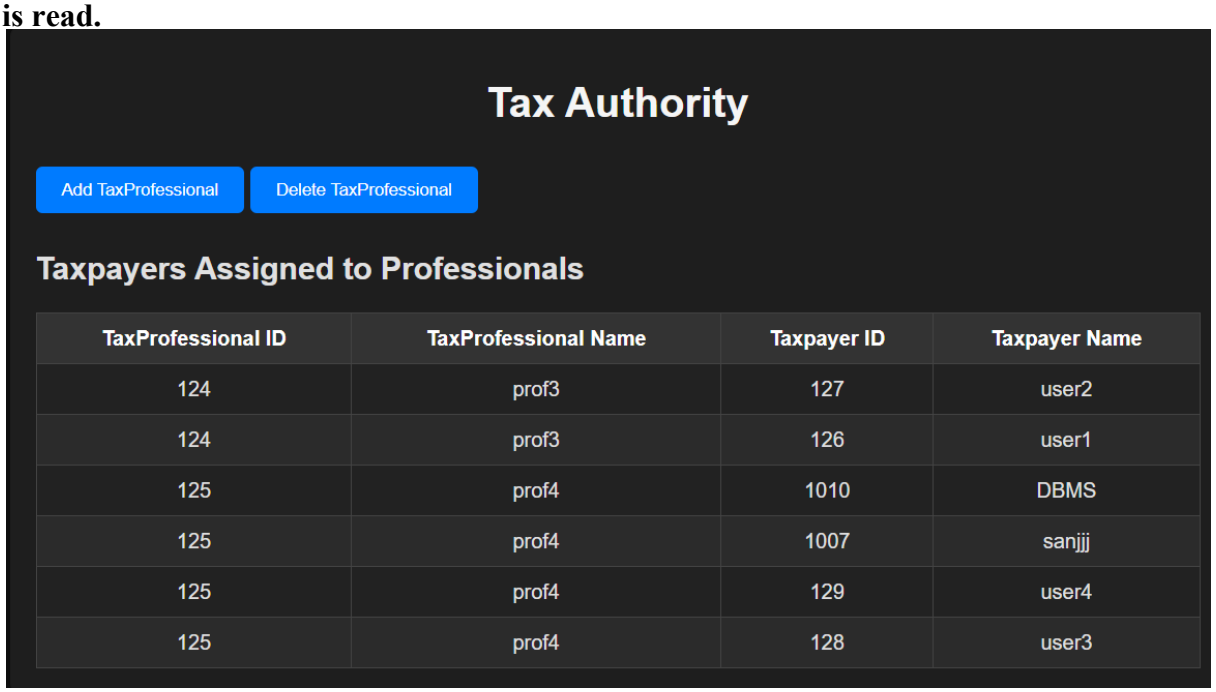


	TaxpayerID	Name	TIN	Address	Email	Phone	RegistrationDate	Password
Edit Copy Delete	16	user1	TIN8381	jnanabharathi	use1@gmail.com	123	2024-11-14	$2y$10$2app4IgnIGXOETmZvMdtOumCfxoygA4L0ZrxxFG6tCh...
Edit Copy Delete	17	user2	TIN6299	jnanabharathi	user2@gmail.com	123	2024-11-14	$2y$10$ALHE9y4fBf/sqIfi3dKm9OoneJJbnhrMTvMsKQJiAdz...
Edit Copy Delete	18	user3	TIN5696	jnanabharathi	user3@gmail.com	123	2024-11-14	$2y$10$TYvtJBTw4yw0tozAYvejMejMUvo.G0GVDUxVecXqsT1...
Edit Copy Delete	19	user4	TIN3697	jnanabharathi	user4@gmail.com	123	2024-11-14	$2y$10$WKGDv3gzHRg0DX8KgqZa3eZCWuLRugdZBWGZS1nqPH9...
Edit Copy Delete	37	sanjjj	TIN5428	sanjjj	sanjjjj@gmail.com	123456789	2024-11-21	$2y$10$B5LPyZkkVeZSDXco8dhQuuAgmTLi5WXbGYSUJcGK3o0...
Edit Copy Delete	38	DBMS	TIN8305	DBMS	DBMA@GMAIL.COM	123456789	2024-12-01	$2y$10$hEmiFMZ8OhtdtDebA4.QF.MePVYotJSdlcN3jHOrABn...

A taxpayer called DBMS is created and the password is hashed for the security and privacy purpose.

READ:

Screenshot of taxauthority table where the data from the taxprofessional and taxpayer is read.



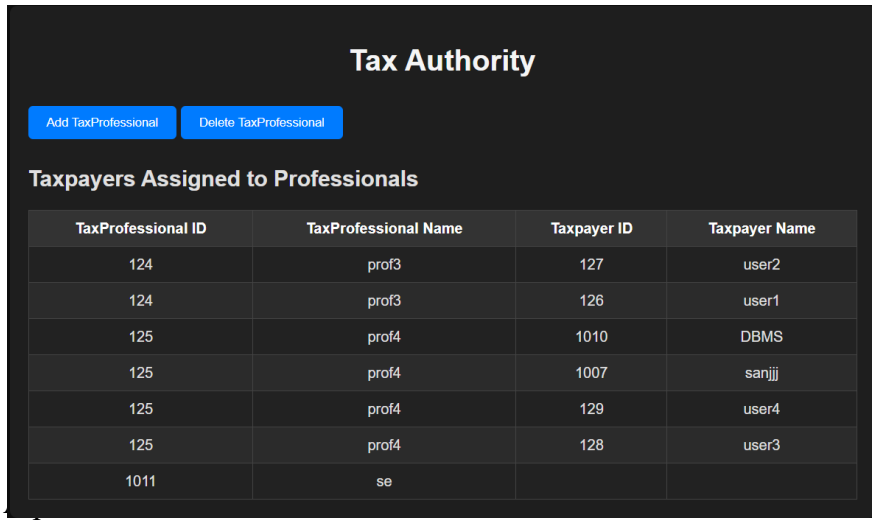
TaxProfessional ID	TaxProfessional Name	Taxpayer ID	Taxpayer Name
124	prof3	127	user2
124	prof3	126	user1
125	prof4	1010	DBMS
125	prof4	1007	sanjjj
125	prof4	129	user4
125	prof4	128	user3

Update:

Screenshot of all files, and the the value recent file will be updated everytime when the user uploads a new file

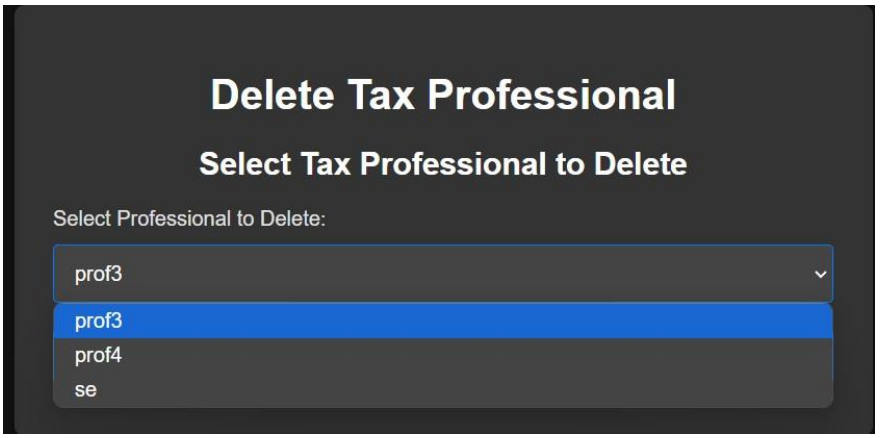
Delete:

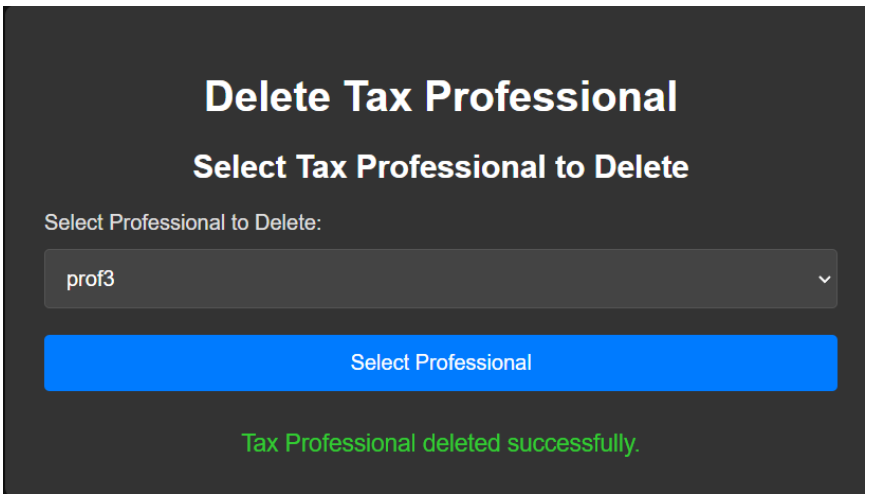
Deleting a taxprofessional from the database

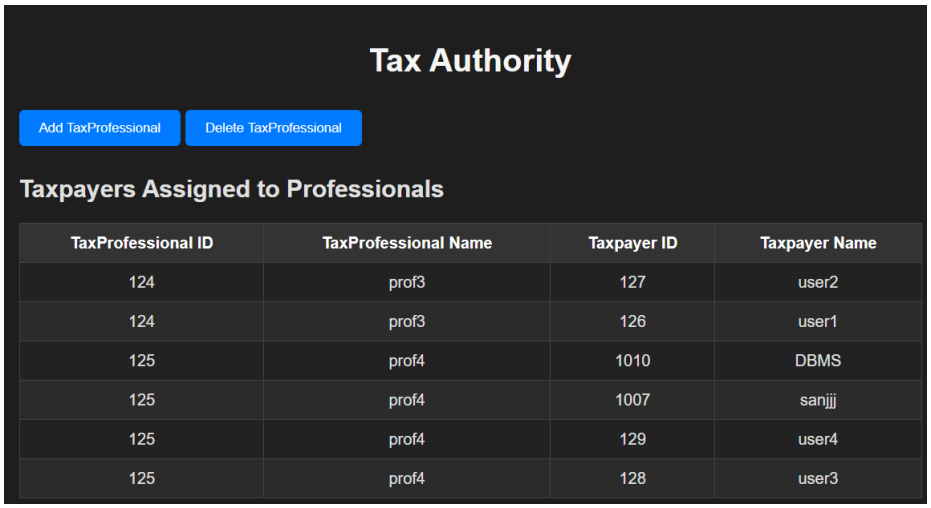


TaxProfessional ID	TaxProfessional Name	Taxpayer ID	Taxpayer Name
124	prof3	127	user2
124	prof3	126	user1
125	prof4	1010	DBMS
125	prof4	1007	sanjjj
125	prof4	129	user4
125	prof4	128	user3
1011	se		

After delete–





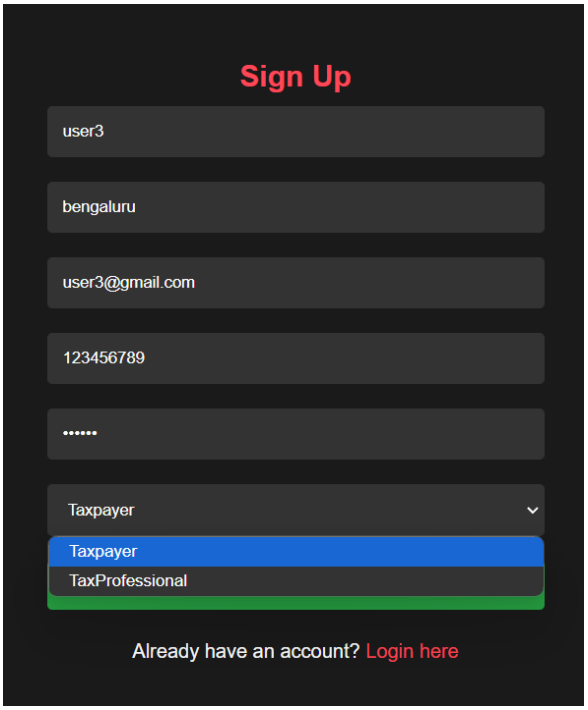


TaxProfessional ID	TaxProfessional Name	Taxpayer ID	Taxpayer Name
124	prof3	127	user2
124	prof3	126	user1
125	prof4	1010	DBMS
125	prof4	1007	sanjjj
125	prof4	129	user4
125	prof4	128	user3

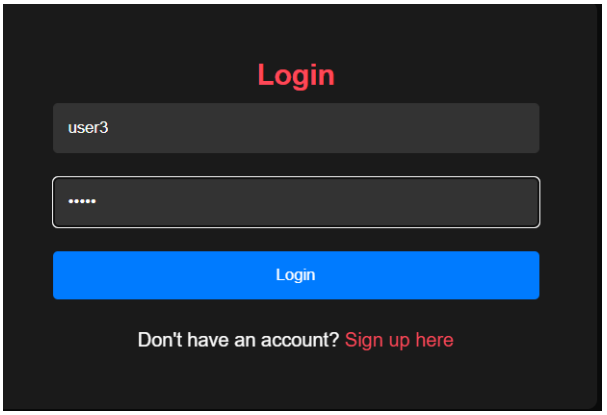
LIST OF FUNCTIONALITIES AND FEATURES:

Feature	Functionalities	Description
Login/Signup	Unified Login & Signup	A single login and signup interface for all three roles: Taxpayer, Tax Professional, and Tax Authority. Where during signup the user can select his role(Taxpayer/Taxprofession al)

Signup



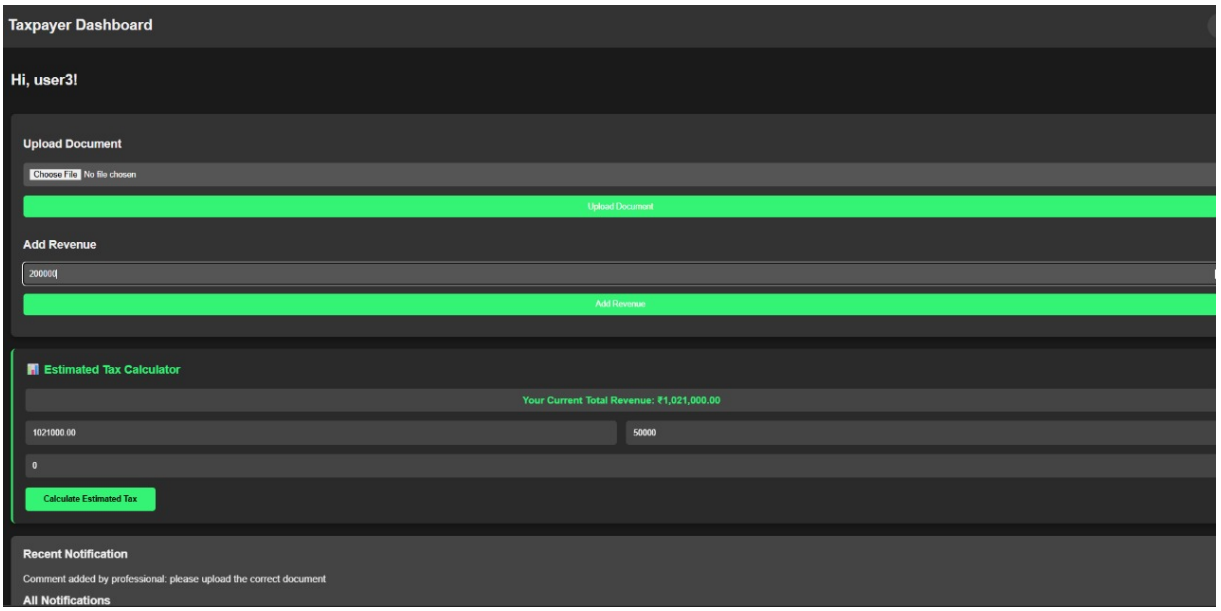
Login:



8.

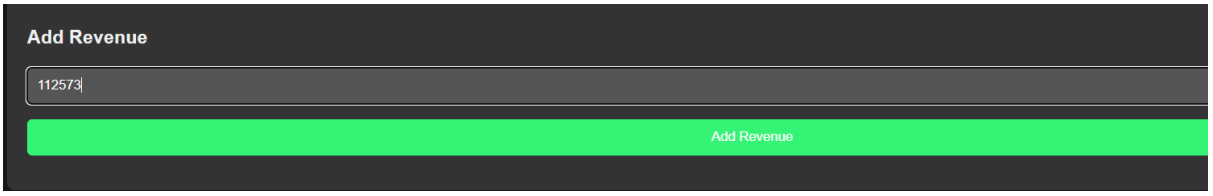
Role	Functionalities	Discription
Taxpayer	Fill Revenue Details	Taxpayers can input their revenue data through an intuitive form.
	Upload Documents	Taxpayers can upload tax-related documents (e.g., invoices, receipts).
	View Notifications	Taxpayers can view personalized notifications sent by their assigned tax professional.

Taxpayer page:

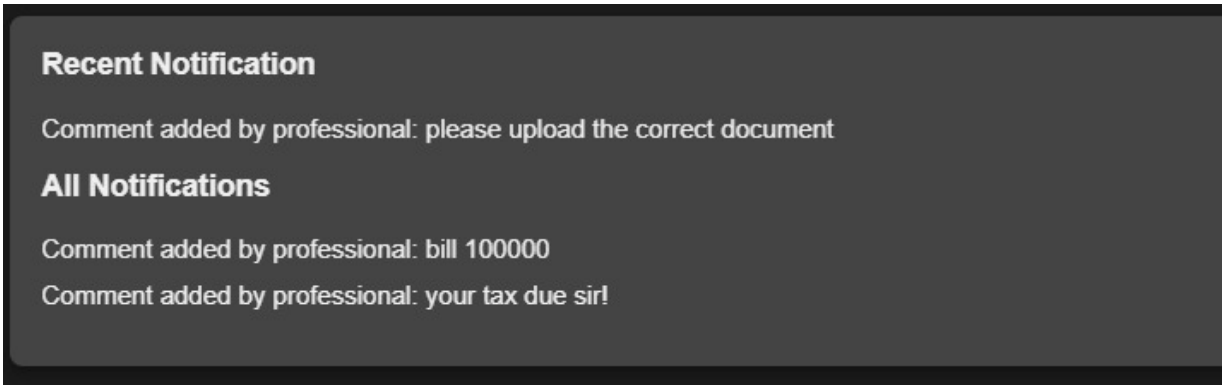


Upload Document:

Revenue:

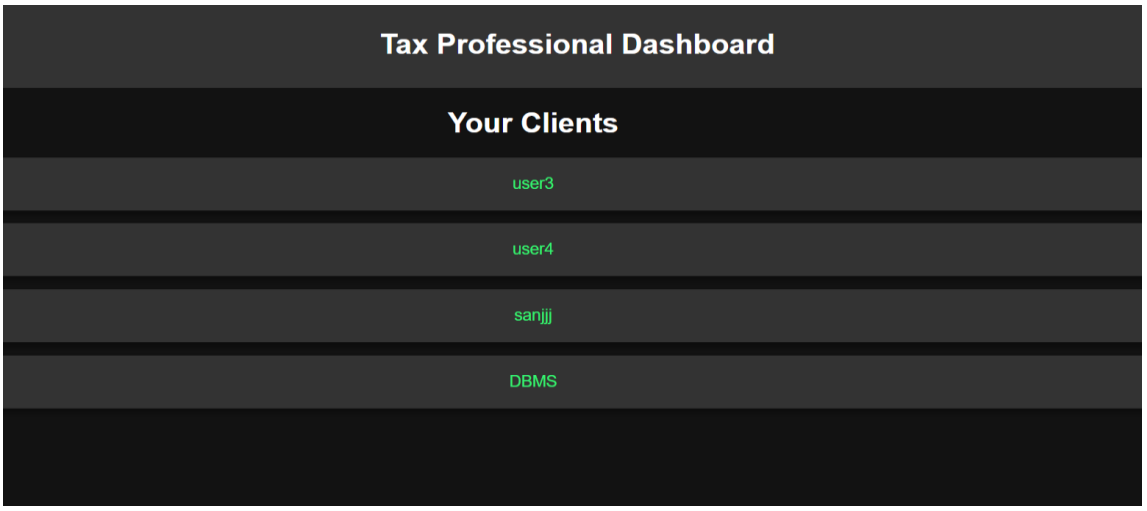


.Notifications:

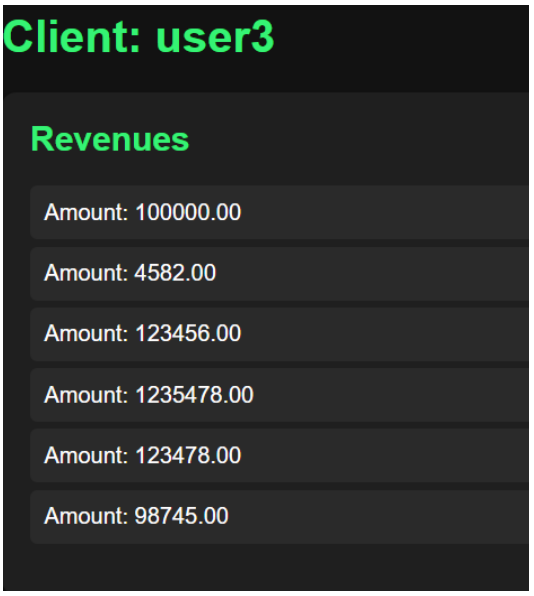


Role	Functionalities	Description
Tax Professional	View Assigned Taxpayers	Tax Professionals can view a list of all taxpayers assigned to them.
	View Revenue Details	Tax Professionals can view the revenue details of each assigned taxpayer individually.
	View Updated Documents	Tax Professionals can view all uploaded files for each taxpayer, including filtering for the most recent ones.
	Send Notifications	Tax Professionals can send custom notifications to their assigned taxpayers.

Taxprofessional dashboard:



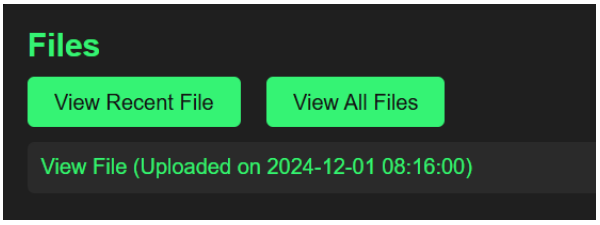
View revenue details:



View uploaded documents:

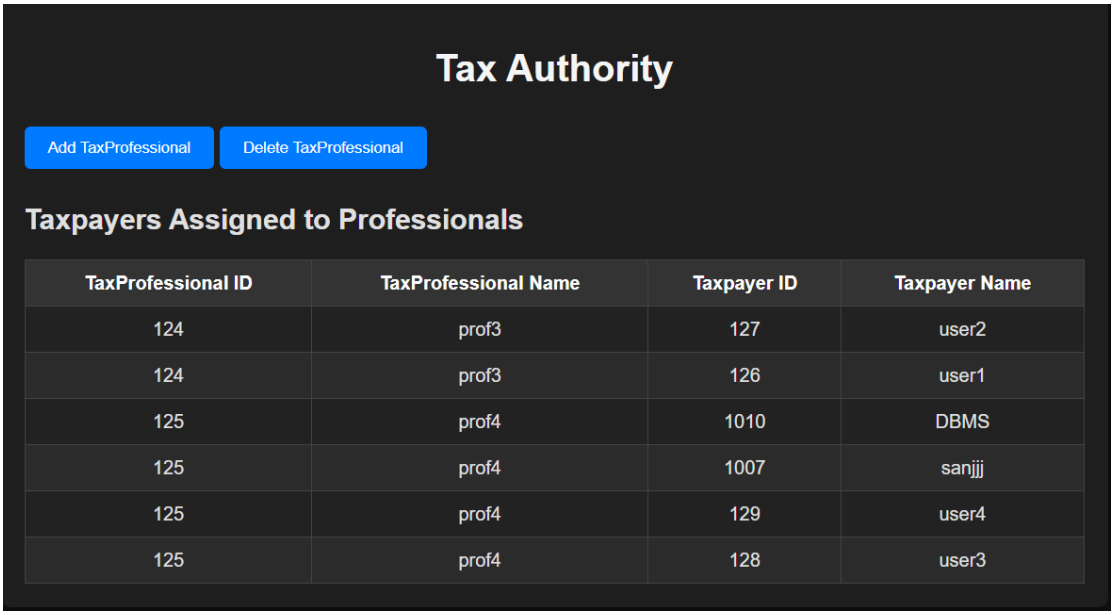
All file→

Recent one→



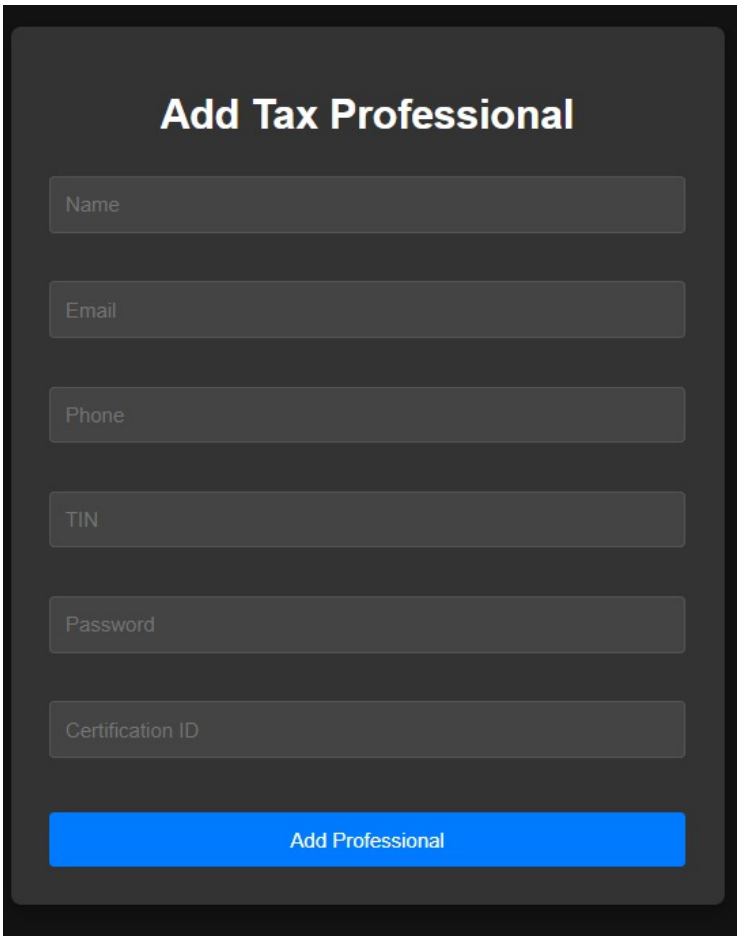
Role	Functionalities	Description
	View Assignments	Tax Authority can view a joined table showing which taxpayer is assigned to which tax professional.
	Add Tax Professionals	Tax Authority can add new tax professionals to the system.
	Delete Tax Professionals	Tax Authority can remove tax professionals, ensuring cascading effects handle dependent data properly.

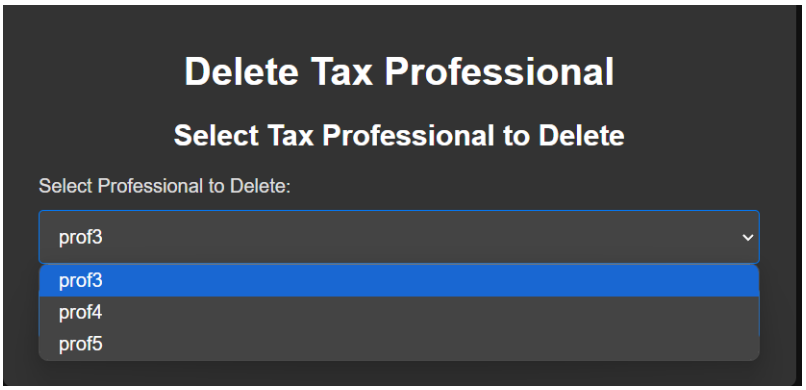
View Assignment table:



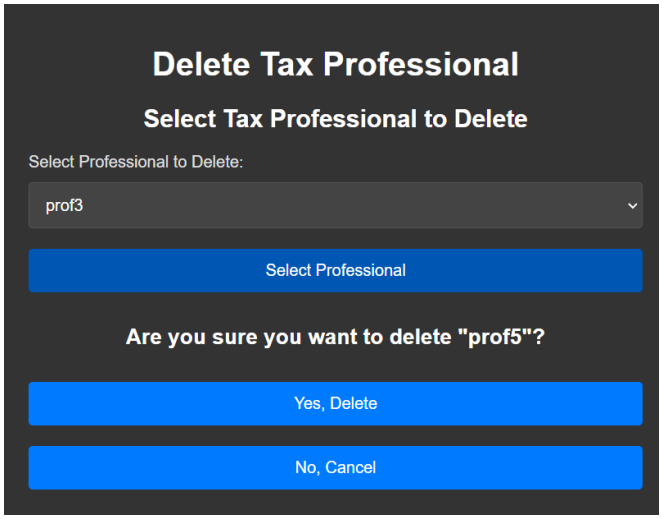
TaxProfessional ID	TaxProfessional Name	Taxpayer ID	Taxpayer Name
124	prof3	127	user2
124	prof3	126	user1
125	prof4	1010	DBMS
125	prof4	1007	sanjjj
125	prof4	129	user4
125	prof4	128	user3

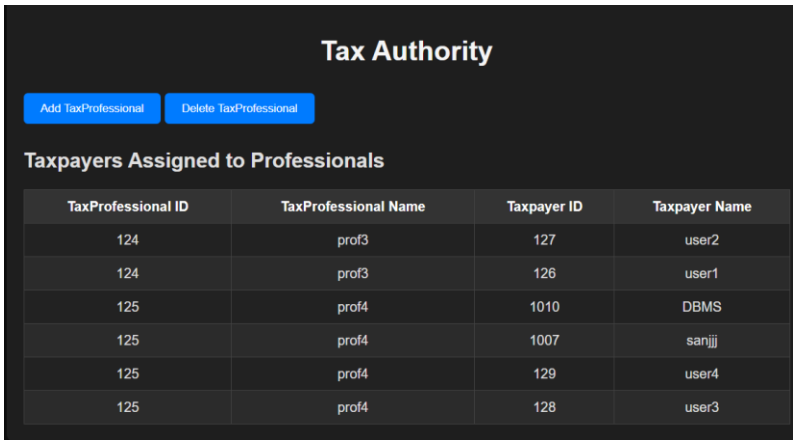
Add Taxprofessional:





Delete professional:





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124	prof3	126	user1
125	prof4	1010	DBMS
125	prof4	1007	sanjjj
125	prof4	129	user4
125	prof4	128	user3

Chapter 6 Conclusion

The Income Tax Management System (ITMS) is a robust, secure, and scalable platform developed to streamline and automate the end-to-end process of income tax administration. It successfully addresses the challenges faced in traditional tax systems by providing a centralized digital solution that supports filing, payment, report generation, and audit logging.

The system supports multiple user roles—Taxpayer, Tax Professional, and Tax Authority—each with designated permissions, ensuring secure and role-based access to sensitive financial data. The integration of stored procedures, triggers, and complex queries enhances system performance, data consistency, and transparency.

By removing manual dependencies, reducing redundancies, and enabling data-driven decisions through comprehensive reporting and analytics, ITMS contributes to improved tax compliance and administrative efficiency. The project also demonstrates strong backend relational database design and application logic, making it adaptable for future enhancements such as AI-powered fraud detection or mobile app integration.

In conclusion, ITMS is not just a project—it is a step forward toward digital governance in public finance, offering a real-world solution that aligns with the evolving needs of modern tax systems.

Chapter 7 Bibliography

1. **.sql file**

   itms.sql

2. **Github repo link**

https://github.com/sadhanashegde/itms